4.3 SPANNING SETS

LINEAR COMBINATION:

If $\underline{w}$ is a vector in a vector space V, then $\underline{w}$ is said to be a "linear combination" of the vectors $\underline{v}_1, \underline{v}_2, \ldots, \underline{v}_r$ in V if $\underline{w}$ can be expressed in the form

$$\underline{w} = k_1\underline{v}_1 + k_2\underline{v}_2 + \cdots + k_r\underline{v}_r$$

where $k_1, k_2, \ldots, k_r$ are scalars. These scalars are called the "coefficients" of the linear combination.

SPAN OF S:

If $S = \{\underline{w}_1, \underline{w}_2, \ldots, \underline{w}_r\}$ is a nonempty set of vectors in a vector space V, then the set W of all possible linear combinations of the vectors in S is a subspace of V. The subspace W is called the subspace of V "spanned" by S. The vectors $\underline{w}_1, \underline{w}_2, \ldots, \underline{w}_r$ in S are said to "span" W, and is written as

$$W = \text{span}\{\underline{w}_1, \underline{w}_2, \ldots, \underline{w}_r\}$$
$$= \text{span}(S)$$

Exercise Set 4.3

Q2. Express the following as linear combinations of $\underline{u} = (2, 1, 4)$, $\underline{v} = (1, -1, 3)$, and $\underline{w} = (3, 2, 5)$.

a. $(-9, -7, -15)$ b. $(6, 11, 6)$

c. $(0, 0, 0)$

Let

$$k_1 \underline{u} + k_2 \underline{v} + k_3 \underline{w} = \underline{r}$$

(Here $\underline{r}$ is assumed any vector of each part)

$$\Rightarrow k_1(2, 1, 4) + k_2(1, -1, 3) + k_3(3, 2, 5) = \underline{r}$$

$$\Rightarrow (2k_1 + k_2 + 3k_3,\ k_1 - k_2 + 2k_3,\ 4k_1 + 3k_2 + 5k_3) = \underline{r}$$

Equating both sides

$$\begin{aligned} 2k_1 + k_2 + 3k_3 &= r_1 \\ k_1 - k_2 + 2k_3 &= r_2 \\ 4k_1 + 3k_2 + 5k_3 &= r_3 \end{aligned}$$

Thus the augmented matrix is

$$\begin{matrix} & & & a & b & c \end{matrix}$$

$$\left[\begin{array}{ccc:ccc} 2 & 1 & 3 & -9 & 6 & 0 \\ 1 & -1 & 2 & -7 & 11 & 0 \\ 4 & 3 & 5 & -15 & 6 & 0 \end{array}\right]$$

$$\left[\begin{array}{ccc:ccc} 1 & -1 & 2 & -7 & 11 & 0 \\ 2 & 1 & 3 & -9 & 6 & 0 \\ 4 & 3 & 5 & -15 & 6 & 0 \end{array}\right] \quad R_{12}$$

$$\left[\begin{array}{ccc:ccc} 1 & -1 & 2 & -7 & 11 & 0 \\ 0 & 3 & -1 & 5 & -16 & 0 \\ 0 & 7 & -3 & 13 & -38 & 0 \end{array}\right] \quad \begin{matrix} R_2 - 2R_1 \\ R_3 - 4R_1 \end{matrix}$$

$$\left[\begin{array}{ccc:ccc} 1 & -1 & 2 & -7 & 11 & 0 \\ 0 & 3 & -1 & 5 & -16 & 0 \\ 0 & 1 & -1 & 3 & -6 & 0 \end{array}\right] \quad R_3 - 2R_2$$

$$\left[\begin{array}{ccc|ccc} 1 & -1 & 2 & -7 & 11 & 0 \\ 0 & 1 & -1 & 3 & -6 & 0 \\ 0 & 3 & -1 & 5 & -16 & 0 \end{array}\right] \quad R_{23}$$

$$\left[\begin{array}{ccc|ccc} 1 & -1 & 2 & -7 & 11 & 0 \\ 0 & 1 & -1 & 3 & -6 & 0 \\ 0 & 0 & 2 & -4 & 2 & 0 \end{array}\right] \quad R_3 - 3R_2$$

$$\left[\begin{array}{ccc|ccc} 1 & -1 & 2 & -7 & 11 & 0 \\ 0 & 1 & -1 & 3 & -6 & 0 \\ 0 & 0 & 1 & -2 & 1 & 0 \end{array}\right] \quad \frac{1}{2}R_3$$

$$\left[\begin{array}{ccc|ccc} 1 & -1 & 0 & -3 & 9 & 0 \\ 0 & 1 & 0 & 1 & -5 & 0 \\ 0 & 0 & 1 & -2 & 1 & 0 \end{array}\right] \quad \begin{array}{l} R_1 - 2R_3 \\ R_2 + R_3 \end{array}$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -2 & 4 & 0 \\ 0 & 1 & 0 & 1 & -5 & 0 \\ 0 & 0 & 1 & -2 & 1 & 0 \end{array}\right] \quad R_1 + R_2$$

Thus

a. $k_1 = -2$, $k_2 = 1$, $k_3 = -2$

b. $k_1 = 4$, $k_2 = -5$, $k_3 = 1$

c. $k_1 = 0$, $k_2 = 0$, $k_3 = 0$

*Verification: [This ~~part~~ is not part of solution].

a. $-2(2,1,4) + (1,-1,3) + -2(3,2,5)$
$= (-4+1-6, -2-1-4, -8+3+-10)$
$= (-9, -7, -15)$

b. $4(2,1,4) - 5(1,-1,3) + (3,2,5)$
$= (8-5+3, 4+5+2, 16-15+5)$
$= (6, 11, 6)$

c. $0(2,1,4) + 0(1,-1,3) + 0(3,2,5)$
$= (0,0,0)$

Q5. In each part, express the vector as a linear combination of

$A=\begin{bmatrix}1 & -1\\ 0 & 2\end{bmatrix}$, $B=\begin{bmatrix}0 & 1\\ 0 & 1\end{bmatrix}$, $C=\begin{bmatrix}0 & 1\\ 0 & 0\end{bmatrix}$, $D=\begin{bmatrix}2 & 0\\ 1 & -1\end{bmatrix}$

b. $\begin{bmatrix}3 & 1\\ 1 & 2\end{bmatrix}$

Let $k_1A + k_2B + k_3C + k_4D = \begin{bmatrix}3 & 1\\ 1 & 2\end{bmatrix}$

or $k_1\begin{bmatrix}1 & -1\\ 0 & 2\end{bmatrix} + k_2\begin{bmatrix}0 & 1\\ 0 & 1\end{bmatrix} + k_3\begin{bmatrix}0 & 1\\ 0 & 0\end{bmatrix} + k_4\begin{bmatrix}2 & 0\\ 1 & -1\end{bmatrix} = \begin{bmatrix}3 & 1\\ 1 & 2\end{bmatrix}$

or $\begin{bmatrix}k_1+0k_2+0k_3+2k_4 & -k_1+k_2+k_3+0k_4\\ 0k_1+0k_2+0k_3+k_4 & 2k_1+k_2+0k_3-k_4\end{bmatrix} = \begin{bmatrix}3 & 1\\ 1 & 2\end{bmatrix}$

Equating both sides

$k_1 + 0k_2 + 0k_3 + 2k_4 = 3$

$-k_1 + k_2 + k_3 + 0k_4 = 1$

$0k_1 + 0k_2 + 0k_3 + k_4 = 1$

$2k_1 + k_2 + 0k_3 - k_4 = 2$

The augmented matrix is

$$\begin{bmatrix}1 & 0 & 0 & 2 & 3\\ -1 & 1 & 1 & 0 & 1\\ 0 & 0 & 0 & 1 & 1\\ 2 & 1 & 0 & -1 & 2\end{bmatrix} \sim \begin{bmatrix}1 & 0 & 0 & 2 & 3\\ 0 & 1 & 1 & 2 & 4\\ 0 & 0 & 0 & 1 & 1\\ 0 & 1 & 0 & -5 & -4\end{bmatrix} \begin{matrix}R_2+R_1\\ \\ \\ R_4-2R_1\end{matrix}$$

$$\sim \begin{bmatrix}1 & 0 & 0 & 2 & 3\\ 0 & 1 & 0 & -5 & -4\\ 0 & 0 & 0 & 1 & 1\\ 0 & 1 & 1 & 2 & 4\end{bmatrix} R_{24} \sim \begin{bmatrix}1 & 0 & 0 & 2 & 3\\ 0 & 1 & 0 & -5 & -4\\ 0 & 1 & 1 & 2 & 4\\ 0 & 0 & 0 & 1 & 1\end{bmatrix} R_{34}$$

$$\sim \begin{bmatrix}1 & 0 & 0 & 2 & 3\\ 0 & 1 & 0 & -5 & -4\\ 0 & 0 & 1 & 7 & 8\\ 0 & 0 & 0 & 1 & 1\end{bmatrix} R_{32} \sim \begin{bmatrix}1 & 0 & 0 & 0 & 1\\ 0 & 1 & 0 & 0 & 1\\ 0 & 0 & 1 & 0 & 1\\ 0 & 0 & 0 & 1 & 1\end{bmatrix} \begin{matrix}R_1-2R_4\\ R_2+5R_4\\ R_3-7R_4\end{matrix}$$

Thus $k_1 = k_2 = k_3 = k_4 = 1$

So that

$$\begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix} + \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 2 & 0 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}$$

Q6. In each part express the vector as a linear combination of $P_1 = 2+x+4x^2$, $P_2 = 1-x+3x^2$, and $P_3 = 3+2x+5x^2$

a. $7+8x+9x^2$

Let $k_1 P_1 + k_2 P_2 + k_3 P_3 = 7+8x+9x^2$

or $k_1(2+x+4x^2) + k_2(1-x+3x^2) + k_3(3+2x+5x^2)$
$= 7+8x+9x^2$

or $(2k_1+k_2+3k_3) + (k_1-k_2+2k_3)x + (4k_1+3k_2+5k_3)x^2$
$= 7+8x+9x^2$

Equating both sides

$$\begin{aligned} 2k_1 + k_2 + 3k_3 &= 7 \\ k_1 - k_2 + 2k_3 &= 8 \\ 4k_1 + 3k_2 + 5k_3 &= 9 \end{aligned}$$

The augmented matrix is

$$\begin{bmatrix} 2 & 1 & 3 & 7 \\ 1 & -1 & 2 & 8 \\ 4 & 3 & 5 & 9 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 2 & 8 \\ 2 & 1 & 3 & 7 \\ 4 & 3 & 5 & 9 \end{bmatrix} R_{12}$$

$$\sim \begin{bmatrix} 1 & -1 & 2 & 8 \\ 0 & 3 & -1 & -9 \\ 0 & 7 & -3 & -23 \end{bmatrix} \begin{matrix} R_2 - 2R_1 \\ R_3 - 4R_1 \end{matrix} \sim \begin{bmatrix} 1 & -1 & 2 & 8 \\ 0 & 3 & -1 & -9 \\ 0 & 1 & -1 & -5 \end{bmatrix} R_{32} - 2R$$

$$\sim \begin{bmatrix} 1 & -1 & 2 & 8 \\ 0 & 1 & -1 & -5 \\ 0 & 3 & -1 & -9 \end{bmatrix} R_{23} \sim \begin{bmatrix} 1 & -1 & 2 & 8 \\ 0 & 1 & -1 & -5 \\ 0 & 0 & 2 & 6 \end{bmatrix} R_3 - 3R_2$$

$$\sim \begin{bmatrix} 1 & -1 & 2 & 8 \\ 0 & 1 & -1 & -5 \\ 0 & 0 & 1 & 3 \end{bmatrix} \frac{1}{2}R_3 \sim \begin{bmatrix} 1 & -1 & 0 & 2 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 3 \end{bmatrix} \begin{matrix} R_1 - 2R_3 \\ R_2 + R_3 \end{matrix}$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 3 \end{bmatrix} R_1 + R_2$$

Thus $k_1 = 0$; $k_2 = -2$; $k_3 = 3$

$0\underline{p}_1 - 2\underline{p}_2 + 3\underline{p}_3 = 7 + 8x + 9x^2$

or $-2(1 - x + 3x^2) + 3(3 + 2x + 5x^2) = 7 + 8x + 9x^2$

Q7. In each part, determine whether the vectors span $\mathbb{R}^3$.

a. $\underline{v}_1 = (2,2,2)$, $\underline{v}_2 = (0,0,3)$, $\underline{v}_3 = (0,1,1)$

b. $\underline{v}_1 = (2,-1,3)$, $\underline{v}_2 = (4,1,2)$, $\underline{v}_3 = (8,-1,8)$

a. Let $k_1\underline{v}_1 + k_2\underline{v}_2 + k_3\underline{v}_3 = (a,b,c) \in \mathbb{R}^3$ arbitrary vector

or $k_1(2,2,2) + k_2(0,0,3) + k_3(0,1,1) = (a,b,c)$

or $(2k_1 + 0k_2 + 0k_3, 2k_1 + 0k_2 + k_3, 2k_1 + 3k_2 + k_3) = (a,b,c)$

By definition of equal vectors

$$2k_1 + 0k_2 + 0k_3 = a$$
$$2k_1 + 0k_2 + k_3 = b$$
$$2k_1 + 3k_2 + k_3 = c$$

The augmented matrix is

$$\begin{bmatrix} 2 & 0 & 0 & a \\ 2 & 0 & 1 & b \\ 2 & 3 & 1 & c \end{bmatrix} \sim \begin{bmatrix} 2 & 0 & 0 & a \\ 0 & 0 & 1 & b-a \\ 0 & 3 & 1 & c-a \end{bmatrix} \begin{matrix} R_2 - R_1 \\ R_3 - R_1 \end{matrix} \sim \begin{bmatrix} 2 & 0 & 0 & a \\ 0 & 0 & 1 & b-a \\ 0 & 3 & 0 & c-b \end{bmatrix} R_3 - R_2$$

$$\sim \begin{bmatrix} 2 & 0 & 0 & a \\ 0 & 3 & 0 & c-b \\ 0 & 0 & 1 & b-a \end{bmatrix} R_{23} \sim \begin{bmatrix} 1 & 0 & 0 & a/2 \\ 0 & 1 & 0 & \frac{c-b}{3} \\ 0 & 0 & 1 & b-a \end{bmatrix} \begin{matrix} \frac{1}{2}R_1 \\ \frac{1}{3}R_2 \end{matrix}$$

Since it is consistent for every vector $(a,b,c) \in \mathbb{R}^3$ so there exist scalars k_1, k_2, k_3 such that

$$k_1\underline{v}_1 + k_2\underline{v}_2 + k_3\underline{v}_3 = (a,b,c)$$

Thus $\underline{v}_1, \underline{v}_2, \underline{v}_3$ spans $\mathbb{R}^3$

b. Consider the matrix of the coefficients of the system

$$k_1\underline{v}_1 + k_2\underline{v}_2 + k_3\underline{v}_3 = (a,b,c)$$

$$A = \begin{bmatrix} 2 & 4 & 8 \\ -1 & 1 & -1 \\ 3 & 2 & 8 \end{bmatrix}$$

$$\det(A) = \begin{vmatrix} 2 & 4 & 8 \\ -1 & 1 & -1 \\ 3 & 2 & 8 \end{vmatrix} \begin{matrix} 2 & 4 \\ -1 & 1 \\ 3 & 2 \end{matrix}$$

$$= (16 + 12 - 16) - (24 - 4 - 32)$$

$$= -12 + 12 = 0$$

Since $\det(A) = 0$, it means that the system of $k_1\underline{v}_1 + k_2\underline{v}_2 + k_3\underline{v}_3 = (a,b,c)$ cannot be consistent for all choices of reals a, b, c. Thus $\underline{v}_1, \underline{v}_2, \underline{v}_3$ does not span $\mathbb{R}^3$.

Q8. Suppose that $\underline{v}_1 = (2,1,0,3)$, $\underline{v}_2 = (3,-1,5,2)$, and $\underline{v}_3 = (-1,0,2,1)$. Which of the following vectors are in the span of $\underline{v}_1, \underline{v}_2, \underline{v}_3$ (Span$\{\underline{v}_1, \underline{v}_2, \underline{v}_3\}$)?

d. $(-4, 6, -13, 4)$

To see that $(-4,6,-13,4)$ belongs to the span$\{\underline{v}_1, \underline{v}_2, \underline{v}_3\}$, it suffices to find scalars k_1, k_2, k_3 such that

$$k_1\underline{v}_1 + k_2\underline{v}_2 + k_3\underline{v}_3 = (-4,6,-13,4)$$

The augmented matrix of the system is

$$\begin{bmatrix} 2 & 3 & -1 & -4 \\ 1 & -1 & 0 & 6 \\ 0 & 5 & 2 & -13 \\ 3 & 2 & 1 & 4 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 0 & 6 \\ 2 & 3 & -1 & -4 \\ 0 & 5 & 2 & -13 \\ 3 & 2 & 1 & 4 \end{bmatrix} R_{12}$$

$$\sim \begin{bmatrix} 1 & -1 & 0 & 6 \\ 0 & 5 & -1 & -16 \\ 0 & 5 & 2 & -13 \\ 0 & 5 & 1 & -14 \end{bmatrix} \begin{matrix} R_2 - 2R_1 \\ R_3 - 3R_1 \\ R_4 - 3R_1 \end{matrix} \sim \begin{bmatrix} 1 & -1 & 0 & 6 \\ 0 & 5 & -1 & -16 \\ 0 & 0 & 3 & 3 \\ 0 & 0 & 2 & 2 \end{bmatrix} \begin{matrix} R_2 - R_4 \\ R_3 - R_4 \\ R_4 - R_4 \end{matrix}$$

$$\sim \begin{bmatrix} 1 & -1 & 0 & 6 \\ 0 & 5 & -1 & -16 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix} \begin{matrix} R_2 \\ R_3 \\ \frac{1}{3}R_3 \\ R_4 \end{matrix} \sim \begin{bmatrix} 1 & -1 & 0 & 6 \\ 0 & 5 & 0 & -15 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{matrix} R_1 + R_2 \\ R_2 \\ R_3 \\ R_4 - R_3 \end{matrix}$$

$$\sim \begin{bmatrix} 1 & -1 & 0 & 6 \\ 0 & 1 & 0 & -3 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{matrix} R_1 + R_2 \\ R_2 \\ R_3 \\ R_4 \end{matrix} \sim \begin{bmatrix} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & -3 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{matrix} R_1 + R_2 \\ R_2 \\ R_3 \\ R_4 \end{matrix}$$

Thus $k_1 = 3, k_2 = -3, k_3 = 1$

As $3(2,1,0,3) - 3(3,-1,5,2) + (-1,0,2,1)$
$= (-4, 6, -13, 4)$

Thus $(-4, 6, -13, 4)$ is in the span of v_1, v_2, and v_3.

Q10. Determine whether the following polynomials span P_2.

$p_1 = 1 + x$, $p_2 = 1 - x$, $p_3 = 1 + x + x^2$, $p_4 = 2 - x^2$

For p_1, p_2, p_3, and p_4 to span P_2, it requires that every vector of P_2 can be expressed as linear combination of p_1, p_2, p_3, and p_4.

i.e. $k_1 p_1 + k_2 p_2 + k_3 p_3 + k_4 p_4 = a + bx + cx^2$
$\forall a, b, c \in R$

The augmented matrix is

$$\begin{bmatrix} 1 & 1 & 1 & 2 & a \\ 1 & -1 & 1 & 0 & b \\ 0 & 0 & 1 & -1 & c \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 & 2 & a \\ 0 & -2 & 0 & -2 & b-a \\ 0 & 0 & 1 & -1 & c \end{bmatrix} R_2 - R_1$$

$$\sim \begin{bmatrix} 1 & 1 & 1 & 2 & a \\ 0 & 1 & 0 & 1 & \frac{a-b}{2} \\ 0 & 0 & 1 & -1 & c \end{bmatrix} \cdot R_2/2$$

This is consistent and does have infinitely many solution (one parameter).

Thus $\underline{p}_1, \underline{p}_2, \underline{p}_3$, and $\underline{p}_4$ spans P_2.

Q1. In each part, determine whether the matrices span M_{22}.

E. $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

For the above vectors to span M_{22}, they must be capable to represent the arbitrary vector $\begin{bmatrix} a & b \\ c & d \end{bmatrix} \in M_{22}$.

i.e. $k_1 \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + k_2 \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} + k_3 \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} + k_4 \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

Because coefficient matrix A is square, it suffices that for vectors to span M_{22}, the coefficient matrix is invertible.

In otherwords, if $\det(A) \neq 0$, they span.

$A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ which is triangular and with all diagonal entries non zero, so $\det(A) \neq 0$ & they do span.